

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2019-2020)

PRE-MEDICAL : ENTHUSIAST COURSE [PHASE-I]

TEST # 12

DATE : 23-10-2019

Test Type : Minor

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry** and **Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
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1. An electron of mass m when accelerated through a potential difference V has de-Broglie wavelength λ . The de-Broglie wavelength associated with a proton of mass M accelerated through the same potential difference will be :-

- (1) $\lambda \frac{m}{M}$ (2) $\lambda \sqrt{\frac{m}{M}}$
(3) $\lambda \frac{M}{m}$ (4) $\lambda \sqrt{\frac{M}{m}}$

2. The kinetic energy of an electron with de-Broglie wavelength of 0.3 nanometer is :-

- (1) 0.168 eV (2) 16.8 eV
(3) 1.68 eV (4) 2.5 eV

3. What is the momentum of a photon having frequency 1.5×10^{13} Hz :-

- (1) 3.3×10^{-29} kg m/s
(2) 3.3×10^{-34} kg m/s
(3) 6.6×10^{-34} kg m/s
(4) 6.6×10^{-30} kg m/s

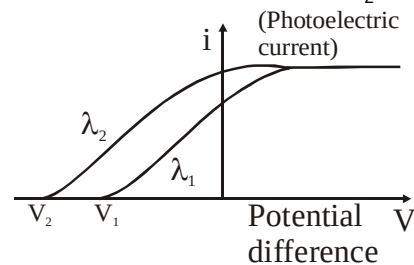
4. When the light source is kept 20 cm away from a photo cell, stopping potential 0.6 V is obtained. When source is kept 40 cm away, the stopping potential will be :-

- (1) 0.3 V (2) 0.6 V
(3) 1.2 V (4) 2.4 V

5. When monochromatic radiation of intensity I falls on a metal surface, the number of photoelectron and their maximum kinetic energy are N and T respectively. If the intensity of radiation is $2I$, the number of emitted electrons and their maximum kinetic energy are respectively :-

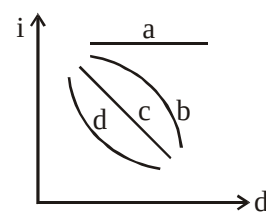
- (1) N and $2T$ (2) $2N$ and T
(3) $2N$ and $2T$ (4) N and T

6. In the following diagram if $V_2 > V_1$ then :-



- (1) $\lambda_1 = \sqrt{\lambda_2}$ (2) $\lambda_1 < \lambda_2$
(3) $\lambda_1 = \lambda_2$ (4) $\lambda_1 > \lambda_2$

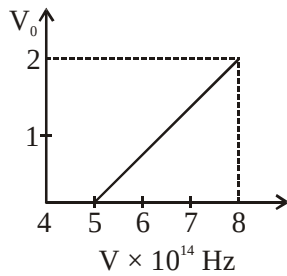
7. A point source of light is used in an experiment on photoelectric effect. Which of the following curves best represents the variation of photo current (i) with distance (d) of the source from the emitter :-



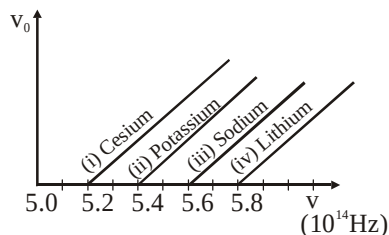
- (1) a (2) b (3) c (4) d

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8. The stopping potential (V_0) versus frequency (ν) plot of a substance is shown in figure the threshold wave length is :-



- (1) 5×10^{14} m
(2) 6000 Å
(3) 5000 Å
(4) Cannot be estimated from given data
9. The figure shows different graphs between stopping potential (V_0) and frequency (ν) for photosensitive surface of cesium, potassium, sodium and lithium. the plots are parallel. Correct ranking of the targets according to their work function greatest first will be :-



- (1) (iv) > (iii) > (ii) > (i)
(2) (i) > (iii) > (ii) > (iv)
(3) (iv) > (iii) > (ii) < (i)
(4) (i) = (iii) = (ii) = (iv)
10. If light of wavelength λ_1 is allowed to fall on a metal, then kinetic energy of photoelectrons emitted is E_1 . If wavelength of light changes to λ_2 then kinetic energy of electrons changes to E_2 . Then work function of the metal is :-

(1) $\frac{E_1 E_2 (\lambda_1 - \lambda_2)}{\lambda_1 \lambda_2}$ (2) $\frac{E_1 \lambda_1 - E_2 \lambda_2}{(\lambda_1 - \lambda_2)}$
(3) $\frac{E_1 \lambda_1 - E_2 \lambda_2}{(\lambda_2 - \lambda_1)}$ (4) $\frac{\lambda_1 \lambda_2 E_1 E_2}{(\lambda_2 - \lambda_1)}$

11. Monochromatic light of wavelength 3000 Å is incident on a surface area 4 cm². If intensity of light is 150 mW/m², then rate at which photons strike the target is :-
(1) 3×10^{10} /s (2) 9×10^{13} /s
(3) 7×10^{15} /s (4) 6×10^{19} /s
12. The minimum intensity of light to be detected by human eye is 10^{-10} W/m². The number of photons of wavelength 5.6×10^{-7} m entering the eye, with pupil area 10^{-6} m², per second for vision will be nearly :-
(1) 100 (2) 200 (3) 300 (4) 400
13. Two identical photo-cathodes receive light of frequencies f_1 and f_2 . If the velocities of the photo electrons (of mass m) coming out are respectively v_1 and v_2 , then :-

(1) $v_1 - v_2 = \left[\frac{2h}{m} (f_1 - f_2) \right]^{1/2}$

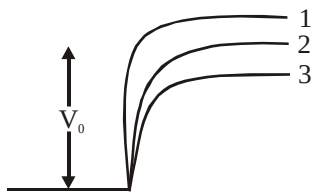
(2) $v_1^2 - v_2^2 = \frac{2h}{m} (f_1 - f_2)$

(3) $v_1 + v_2 = \left[\frac{2h}{m} (f_1 + f_2) \right]^{1/2}$

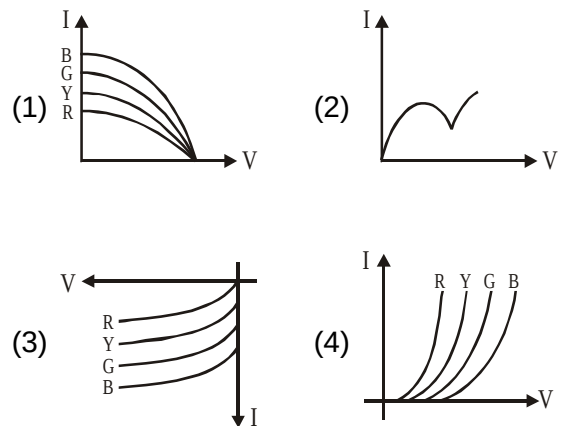
(4) $v_1^2 + v_2^2 = \frac{2h}{m} (f_1 + f_2)$

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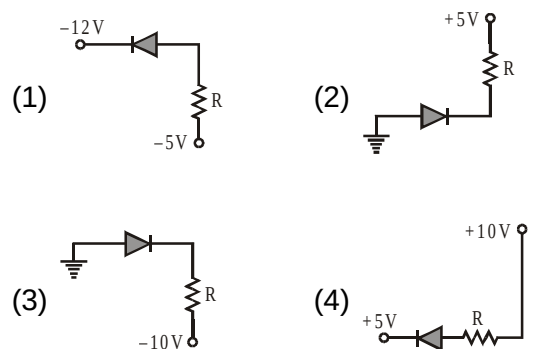
14. When radiation of wavelength λ is incident on a metallic surface, the stopping potential is 4.8 volts. If the same surface is illuminated with radiation of double the wavelength, then the stopping potential becomes 1.6 volts. Then the threshold wavelength for the surface is :-
 (1) 2λ (2) 4λ (3) 6λ (4) 8λ
15. If a source of power 4kW produces 10^{20} photons/second, the radiation belongs to part of the spectrum called :-
 (1) γ -rays (2) X-rays
 (3) Ultraviolet rays (4) Microwaves
16. Photoelectric effect supports quantum nature of light because :-
 (i) There is minimum frequency of light below which no photoelectrons are emitted.
 (ii) Electric charge of photoelectrons is quantized.
 (iii) Maximum kinetic energy of photoelectrons depends only on the frequency of light and not on its intensity.
 (iv) Even when metal surface is faintly illuminated the photoelectrons leave the surface immediately.
 (1) (i), (ii), (iii) (2) (i), (ii), (iv)
 (3) (ii), (iii), (iv) (4) (i), (iii), (iv)
17. In Figure, V_0 is the potential barrier across a p-n junction :-



- (1) 1 and 3 both correspond to forward bias of junction
 (2) 3 corresponds to forward bias of junction and 1 corresponds to reverse bias of junction
 (3) 1 corresponds to forward bias and 3 corresponds to reverse bias of junction
 (4) 3 and 1 both correspond to reverse bias of junction
18. The I-V characteristic of an LED is :-
 R : Red, Y : Yellow, G : Green, B : Blue

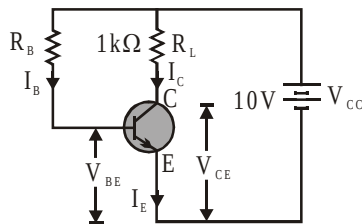


19. Of the diodes shown in the following diagrams, which one is reverse biased ?



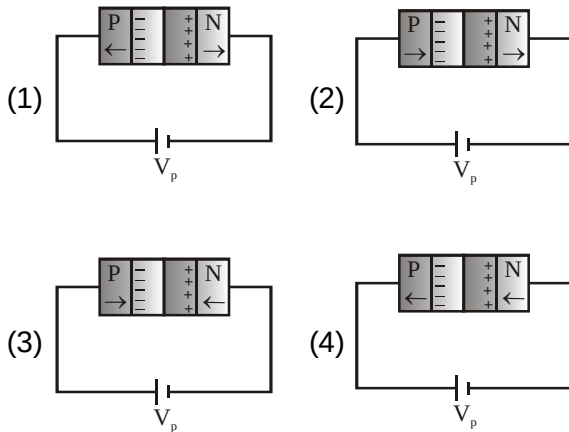
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20. In the circuit shown in Figure, the current gain, $\beta = 100$ for the transistor. What would be the bias resistance R_B so that $V_{CE} = 5V$? (Neglect V_{BE}).

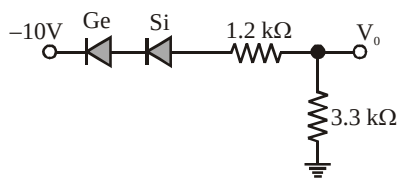


- (1) $2 \times 10^3 \Omega$ (2) $2 \times 10^5 \Omega$
(3) $1 \times 10^6 \Omega$ (4) 500Ω

21. In the case of forward biasing of PN-junction, which one of the following figures correctly depicts the direction of flow of carriers :-

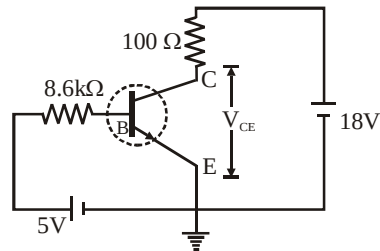


22. Determine V_0 for the network

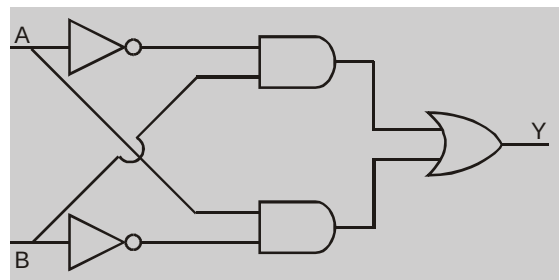


- (1) 6.6 V (2) 9V
(3) - 9V (4) - 6.6 V

23. For the transistor circuit shown below, if $\beta = 100$. Voltage drop between emitter and base is 0.7 V then value of V_{CE} will be :-



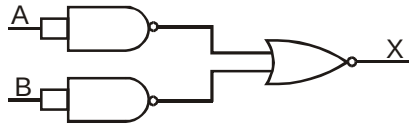
- (1) 10 V (2) 5 V (3) 13 V (4) 0 V
24. The electrical conductivity of a semiconductor increases when electromagnetic radiation of wavelength shorter than 2480 nm is incident on it. Find the band gap of the semiconductor. Given : $h = 6.63 \times 10^{-34} \text{ J-s}$, $c = 3.0 \times 10^8 \text{ ms}^{-1}$ and $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$
(1) 0.8 eV (2) 0.5 eV
(3) 0.3 eV (4) 1.4 eV
25. The combination of the gates shown will produce



- (1) AND gate (2) NAND gate
(3) NOR gate (4) XOR gate

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26. The truth table for the gate shown in the following figure will be



(1)

A	B	X
0	0	0
0	1	1
1	0	1
1	1	0

(2)

A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

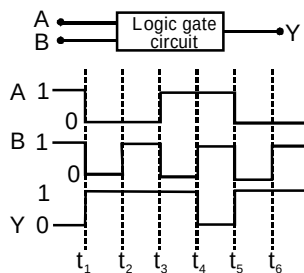
(3)

A	B	X
0	0	1
0	1	1
1	0	1
1	1	0

(4)

A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

27. The following figure shows a logic gate circuit with two inputs A and B and the output Y. The voltage waveforms of A, B, and Y are as given:-



The logic gate is :-

- (1) OR (2) AND
(3) NAND (4) NOR

28. The band gap in Germanium and silicon in eV respectively is

- (1) 0.7, 1.1 (2) 1.1, 0.7
(3) 1.1, 0 (4) 0, 1.1

29. P-type semiconductors are made by adding impurity element

- (1) As (2) P
(3) B (4) Bi

30. At room temperature, a P-type semiconductor has

- (1) Large number of holes and few electrons
(2) Large number of free electrons and few holes
(3) Equal number of free electrons and holes
(4) No electrons or holes

31. In intrinsic semiconductor at room temperature, number of electrons and holes are

- (1) Unequal (2) Equal

- (3) Infinite (4) Zero

32. The valence band and conduction band of a solid overlap at low temperature, the solid may be

- (1) A metal
(2) A semiconductor
(3) An insulator
(4) None of these

33. Which impurity is doped in Si to form N-type semi-conductor?

- (1) Al (2) B
(3) As (4) None of these

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| <p>34. In a semiconductor
 (1) There are no free electrons at any temperature
 (2) The number of free electrons is more than that in a conductor
 (3) There are no free electrons at 0 K
 (4) None of these</p> <p>35. The energy band gap is maximum in
 (1) Metals
 (2) Superconductors
 (3) Insulators
 (4) Semiconductors</p> <p>36. The ionization potential for second He electron is
 (1) 13.6 eV (2) 27.2 eV
 (3) 54.4 eV (4) 100 eV</p> <p>37. The energy required to remove an electron in a hydrogen atom from ground state is
 (1) 13.6 eV (2) 1.36 eV
 (3) 0.136 eV (4) 0.0136 eV</p> <p>38. Every series of hydrogen spectrum has an upper and lower limit in wavelength. The spectral series which has an upper limit of wavelength equal to $18752 \text{ \AA}$ is
 (1) Balmer series (2) Lyman series
 (3) Paschen series (4) Pfund series</p> <p>39. Ionization potential of hydrogen atom is 13.6 V.
 Hydrogen atoms in the ground state are excited by monochromatic radiation of photon energy 12.1 eV. The spectral lines emitted by hydrogen atoms according to Bohr's theory will be
 (1) One (2) Two
 (3) Three (4) Four</p> | <p>40. The ratio of the energies of the hydrogen atom in its first to second excited state is
 (1) $1/4$ (2) $4/9$ (3) $9/4$ (4) 4</p> <p>41. The ionisation potential of hydrogen atom is 13.6 volt. The energy required to remove an electron in the $n = 2$ state of the hydrogen atom is
 (1) 27.2 eV (2) 13.6 eV
 (3) 6.8 eV (4) 3.4 eV</p> <p>42. In an amplitude modulated wave for audio frequency of 500 cycle/second, the appropriate carrier frequency will be
 (1) 50 cycles/sec
 (2) 100 cycles/sec
 (3) 500 cycles/sec
 (4) 50,000 cycles/sec</p> <p>43. Through which mode of propagation, the radio waves can be sent from one place to another
 (1) Ground wave propagation
 (2) Sky wave propagation
 (3) Space wave propagation
 (4) All of them</p> <p>44. 17. The radio waves of frequency 300 MHz to 3000 MHz belong to
 (1) High frequency band
 (2) Very high frequency band
 (3) Ultra high frequency band
 (4) Super high frequency band</p> <p>45. The carrier frequency generated by a tank circuit containing 1 nF capacitor and 10 H inductor is
 (1) 1592 Hz (2) 1592 MHz
 (3) 1592 kHz (4) 159.2 Hz</p> |
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46. For a catalyst which condition is not essential :
- Variable valency
 - High ionisation energy
 - Empty orbitals
 - Free valency on the surface
47. Density of which of the following element is highest :
- Pt
 - Hg
 - Mn
 - Cu
48. $\text{Cr}_2\text{O}_7^{2-} \xrightleftharpoons[X]{Y} 2\text{CrO}_4^{2-}$, X and Y are respectively :
- X = OH⁻, Y = H⁺
 - X = H⁺, Y = OH⁻
 - X = OH⁻, Y = H₂O₂
 - X = H₂O₂, Y = OH⁻
49. $\text{Cu} + \text{conc. HNO}_3 \longrightarrow \text{Cu}(\text{NO}_3)_2 + \text{X}$
(hot)
(oxide of nitrogen); then X is :
- N₂O
 - NO₂
 - NO
 - N₂O₃
50. Copper becomes green when exposed to moist air for longer period.
- Because of the formation of a layer of cupric oxide on the surface of copper
 - Because of the formation of a layer of basic carbonate of copper on the surface of copper
 - Because of the formation of a layer of cupric hydroxide on the surface of copper.
 - (1) and (3) both
51. Cl₂ gas is obtained by various reactions but not by :
- KMnO₄ + conc. HCl $\xrightarrow{\Delta}$
 - KCl + K₂Cr₂O₇ + conc. H₂SO₄ $\xrightarrow{\Delta}$
 - MnO₂ + conc. HCl $\xrightarrow{\Delta}$
 - KCl + F₂ $\xrightarrow{\Delta}$
52. Maximum magnetic moment is shown by :
- d⁵
 - d⁶
 - d⁷
 - d⁸
53. Which of the following reactions is used to estimate copper volumetrically ?
- $2\text{Cu}^{2+} + 4\text{CN}^- \longrightarrow \text{Cu}_2(\text{CN})_2 + (\text{CN})_2$
 - $\text{Cu}^{2+} + 4\text{NH}_3 \longrightarrow [\text{Cu}(\text{NH}_3)_4]^{2+}$
 - $2\text{Cu}^{2+} + 2\text{CNS}^- + \text{SO}_2 + 2\text{H}_2\text{O} \longrightarrow \text{Cu}_2(\text{CNS})_2 + \text{H}_2\text{SO}_4 + 2\text{H}^+$
 - $2\text{Cu}^{2+} + 4\text{I}^- \longrightarrow \text{Cu}_2\text{I}_2 + \text{I}_2$
54. Which of the following compounds is most sensitive of light ?
- AgCl
 - AgCN
 - AgI
 - AgBr
55. (NH₄)₂Cr₂O₇ (Ammonium dichromate) is used in fire works. The green coloured powder blown in air is :
- Cr₂O₃
 - CrO₂
 - Cr₂O₄
 - CrO₃
56. Which of the following chemical reaction(s) is (are) involved in the developing of photographic plate ?
- $\text{C}_6\text{H}_4(\text{OH})_2 + 2\text{AgCl} \longrightarrow 2\text{Ag} + \text{C}_6\text{H}_4\text{O}_2 + 2\text{HCl}$
 - $\text{AgBr} + 2\text{Na}_2\text{S}_2\text{O}_3 \longrightarrow \text{Na}_3[\text{AgS}_2\text{O}_3] + \text{NaBr}$
 - $\text{AgBr} + 2\text{NH}_3(\text{aq}) \longrightarrow [\text{Ag}(\text{NH}_3)_2] \text{Br}$
 - $2\text{AgBr} + \text{Na}_2\text{S}_2\text{O}_3 \longrightarrow \text{Ag}_2\text{S}_2\text{O}_3 + 2\text{NaBr}$

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| 57. The number of P—O—P bonds in cyclic metaphosphoric acid is –
(1) Zero (2) Two
(3) Three (4) Four | 64. Name of the structure of silicates in which three oxygen atoms of $[\text{SiO}_4]^{4-}$ are shared is –
(1) Pyrosilicate
(2) Sheet silicate
(3) Linear chain silicate
(4) Three dimensional silicate |
| 58. Amongst H_2O , H_2S , H_2Se and H_2Te , the one with the highest boiling point is –
(1) H_2O because of hydrogen bonding
(2) H_2Te because of higher molecular weight
(3) H_2S because of hydrogen bonding
(4) H_2Se because of lower molecular weight. | 65. $\text{B(OH)}_3 + \text{NaOH} \rightleftharpoons \text{NaBO}_2 + \text{Na[B(OH)}_4] + \text{H}_2\text{O}$
how can this reaction is made to proceed in forward direction ?
(1) Addition of cis 1, 2 diol
(2) Addition of borax
(3) Addition of trans 1, 2 diol
(4) Addition of Na_2HPO_4 |
| 59. Which of the following are hydrolysed –
(1) NCl_3 (2) SiCl_4
(3) CCl_4 (4) Both (1) & (2) | 66. A solid has a structure in which W atoms are located at the corners of a cubic lattice, O atom at the centre of the edges and Na atom at centre of the cubic. The formula for the compound is :-
(1) NaWO_2 (2) NaWO_3
(3) Na_2WO_3 (4) NaWO_4 |
| 60. The set with correct order of acidity is –
(1) $\text{HClO} < \text{HClO}_2 < \text{HClO}_3 < \text{HClO}_4$
(2) $\text{HClO}_4 < \text{HClO}_3 < \text{HClO}_2 < \text{HClO}$
(3) $\text{HClO} < \text{HClO}_4 < \text{HClO}_3 < \text{HClO}_2$
(4) $\text{HClO}_4 < \text{HClO}_2 < \text{HClO}_3 < \text{HClO}$ | 67. Which of the following statements is correct in the rock-salt structure of an ionic compounds?
(1) coordination number of cation is four whereas that of anion is six.
(2) coordination number of cation is six whereas that of anion is four.
(3) coordination number of each cation and anion is four.
(4) coordination number of each cation and anion is six. |
| 61. The number of S—S bonds in sulphur trioxide trimer, (S_3O_9) is –
(1) Three (2) Two
(3) One (4) Zero | |
| 62. H_3BO_3 is –
(1) Monobasic acid and weak Lewis acid
(2) Monobasic and weak Bronsted acid
(3) Monobasic and strong Lewis acid
(4) Tribasic and weak Bronsted acid | |
| 63. Which is the most thermodynamically stable allotropic form of phosphorus?
(1) Red (2) White
(3) Black (4) Yellow | |

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| <p>68. The coordination number of cation and anion in Fluorite CaF_2 and CsCl are respectively :-
 (1) 8 : 4 and 6 : 3 (2) 6 : 3 and 4 : 4
 (3) 8 : 4 and 8 : 8 (4) 4 : 2 and 2 : 4</p> <p>69. The tetrahedral voids formed by ccp arrangement of Cl^- ions in rock salt structure are :-
 (1) occupied by Na^+ ions
 (2) occupied by Cl^- ions
 (3) occupied by either Na^+ or Cl^- ions
 (4) vacant</p> <p>70. The number of nearest neighbours around each particle in a face-centred cubic lattice is :-
 (1) 4 (2) 6 (3) 8 (4) 12</p> <p>71. In the closest packing of atoms A (radius : r_a), the radius of atom B that can be fitted into tetrahedral voids is :-
 (1) $0.155 r_a$ (2) $0.225 r_a$
 (3) $0.414 r_a$ (4) $0.732 r_a$</p> <p>72. The oxidation potential of a hydrogen electrode at pH = 10 and $P_{\text{H}_2} = 1$ is :-
 (1) 0.51 V (2) 0.00 V
 (3) + 0.59 V (4) 0.059 V</p> <p>73. KF crystallizes in the NaCl type structure. If the radius of K^+ ions 132 pm and that of F^- ion is 135 pm, what is the closet K-K distance ?
 (1) Can not say (2) 534 pm
 (3) 755 pm (4) 378 pm</p> <p>74. $r_{\text{Na}^+} = 95$ pm and $r_{\text{Cl}^-} = 181$ pm in NaCl (rock salt) structure. What is the shortest distance between Na^+ ions ?
 (1) 778.3 pm (2) 276 pm
 (3) 195.7 pm (4) 390.3 pm</p> <p>75. In diamond, carbon atom occupy FCC lattice points as well as alternate tetrahedral voids. If edge length of the unit cell is 356 pm, then radius of carbon atom is :-</p> | <p>(1) 77.07 pm (2) 154.14 pm
 (3) 251.7 pm (4) 89 pm</p> <p>76. Give the correct order of initials T (true) or F (False) for following statements :
 I. In an anti-fluorite structure anions form FCC lattice and cations occupy all tetrahedral voids.
 II. If the radius of cations and anions are $0.2\text{\AA}$ and $0.95\text{\AA}$ then coordinate number of cation in the crystal is 4.
 III. An atom/ion is transferred from a lattice site to an interstitial position in Frenkel defect.
 IV. Density of crystal always increases due to substitutinal impurity defect.
 (1) TFFF (2) FTTF
 (3) TFFT (4) TFTF</p> <p>77. Xenon crystallises in the face-centred cubic lattice and the edge of the unit cell is 620 pm. What is the next nearest neighbour distance ?
 (1) 738.5 pm (2) 620 pm
 (3) 438.5 pm (4) 310 pm</p> <p>78. An element crystallizes in a structure having FCC unit cell of an edge 200 pm. Calculate the density, if 200 g of this element contains 5×10^{24} atoms.
 (1) 5 m/cm^3 (2) 30 gm/cm^3
 (3) 10 gm/cm^3 (4) 20 gm/cm^3</p> <p>79. If a spoon of copper metal is placed in a solution of ferrous sulphate :-
 (1) Cu will precipitate out
 (2) iron will precipitate
 (3) Cu and Fe will precipitate
 (4) no reaction will take place</p> |
|---|--|

SPACE FOR ROUGH WORK

80. During discharge of a lead storage cell the density of sulphuric acid in the cell :-
 (1) increasing
 (2) decreasing
 (3) remains unchanged
 (4) initially increases but decreases subsequently
81. If the solution of the CuSO_4 in which copper rod is immersed is diluted to 10 times, the electrode reduction potential. ($T = 298 \text{ K}$) :-
 (1) increases by 0.030 volt
 (2) decreases by 0.030 volt
 (3) increases by 0.059 volt
 (4) decreases by 0.0059 volt
82. The reduction potential values are given below
 $\text{Al}^{3+} / \text{Al} = -1.67 \text{ volt}$, $\text{Mg}^{2+} / \text{Mg} = -2.34 \text{ volt}$
 $\text{Cu}^{2+} / \text{Cu} = +0.34 \text{ volt}$
 $\text{I}_2 / 2\text{I}^- = +0.53 \text{ volt}$. Which one is the best reducing agent ?
 (1) Al (2) Mg (3) Cu (4) I_2
83. Given standard electrode potentials :
 $\text{Fe}^{3+} + 3\text{e}^- \longrightarrow \text{Fe}$; $E^\circ = -0.036 \text{ volt}$
 $\text{Fe}^{2+} + 2\text{e}^- \longrightarrow \text{Fe}$; $E^\circ = -0.440 \text{ volt}$
 The standard electrode potential E° for $\text{Fe}^{3+} + \text{e}^- \longrightarrow \text{Fe}^{2+}$ is :-
 (1) -0.476 volt (2) -0.404 volt
 (3) 0.440 volt (4) 0.772 volt
84. When the electric current is passed through a cell having an electrolyte, the positive ions move towards cathode and negative ions towards the anode. If the cathode is pulled out of the solution :-
 (1) the positive and negative ions will move towards anode
 (2) the positive ions will start moving towards the anode while negative ions will stop moving
 (3) the negative ions will continue to move towards anode while positive ions will stop moving
 (4) the positive and negative ions will start moving randomly
85. The oxidation potentials of Zn, Cu, Ag, H_2 and Ni are 0.76, -0.34, -0.80, 0.00, 0.25 volt respectively. Which of the following reactions will provide maximum voltage :-
 (1) $\text{Zn} + \text{Cu}^{2+} \longrightarrow \text{Cu} + \text{Zn}^{2+}$
 (2) $\text{Zn} + 2\text{Ag}^+ \longrightarrow 2\text{Ag} + \text{Zn}^{2+}$
 (3) $\text{H}_2 + \text{Cu}^{2+} \longrightarrow 2\text{H}^+ + \text{Cu}$
 (4) $\text{H}_2 + \text{Ni}^{2+} \longrightarrow 2\text{H}^+ + \text{Ni}$
86. How many farads of electricity is required to reduce one mol of MnO_4^- to Mn^{2+} :-
 (1) 1 (2) 2 (3) 3 (4) 5
87. $\text{Zn} | \text{Zn}^{2+} (C_1) || \text{Zn}^{2+} (C_2) | \text{Zn}$. for this cell ΔG is negative if :-
 (1) $C_1 = C_2$ (2) $C_1 > C_2$ (3) $C_2 > C_1$ (4) None
88. $\text{Pt} \left| \text{H}_2 \right| \left| \text{H}^+ \right| \left| \text{H}^+ \right| \left| \text{H}_2 \right| \text{Pt}$ (where p_1 and p_2 are pressure) cell reaction will be spontaneous if :-
 (1) $p_1 = p_2$ (2) $p_1 > p_2$
 (3) $p_2 > p_1$ (4) $p_1 = 1 \text{ atm}$
89. $\text{M}^{2+} + 2\text{e}^- \longrightarrow \text{M}$. 0.275 g of metal M is deposited at the cathode due to passage of 1A of current for 965 s. Hence atomic weight of the metal M is :-
 (1) 27.5 (2) 55.0 (3) 110.0 (4) 13.75
90. If same quantity of electricity is passed through CuCl and CuSO_4 , the ratio of the weights of Cu deposited from CuSO_4 and CuCl is :-
 (1) 2 : 1 (2) 1 : 2 (3) 1 : 1 (4) 4 : 1

SPACE FOR ROUGH WORK

91. (A) Pollen grain are generally spherical measuring about 25-50 μ m in diameter
(B) In over 60% of angiosperms pollen grains are shed at 2 celled stage
(1) Both (A) and (B) are correct
(2) Only (B) is correct
(3) Only (A) is correct
(4) Both (A) and (B) are incorrect
92. Pick the wrong one family with respect to pollen viability in months :-
(1) Rosaceae (2) Solanaceae
(3) Leguminosae (4) Cucurbitaceae
93. Which one of the following statement is wrong?
(1) Some extent of pollen viability depends on prevailing temperature and humidity
(2) In *Michelia* gynoecium is multicarpellary apocarpous
(3) In *papaver* pistil is multicarpellary syncarpous
(4) Ovary of wheat, paddy and orchid has single ovule
94. Select the odd one about ploidy level :-
(1) Nucellus (2) Endosperm
(3) Perisperm (4) Integument
95. A typical angiosperm embryo sac, at maturity.
(1) 7 celled + 8 nucleate
(2) 6 celled + 7 nucleate
(3) 4 celled + 4 nucleate
(4) 8 celled + 7 nucleate
96. (A) Polar nuclei are situated below the egg apparatus in the large central cell
(B) Filiform apparatus play important role in guiding the pollen tubes into ovule
(1) Both (A) and (B) are correct
(2) Both (A) and (B) are incorrect
(3) Only (A) is correct
(4) Only (B) is correct
97. Select wrong one with respect to cleistogamous flower.
(1) They are invariably autogamous
(2) They produce assured seed set even in the absence of pollinators
(3) Found in *viola*, *Oxalis* and *Commelina*
(4) Found in all dichogamous plants
98. What are the number of meiotic division and mitotic divisions required for the development of four male gametophyte from PMCs respectively?
(1) 1, 2 (2) 3, 6 (3) 1, 8 (4) 2, 2
99. Features like large feathery stigma, non sticky pollen grains and single ovule in each ovary are characteristics of plants pollinated by :-
(1) Insect (2) Water
(3) Air (4) Birds
100. Exine is made up of :-
(1) Sporopollenin (2) Pectin
(3) Cellulose (4) Lipids
101. Select the number of organisms from given list, that show triple fusion :-
Ficus, *Marchantia*, *Cononut*, Bean, *Cycas*, Castor, Sweet potato.
(1) Four (2) Six
(3) Five (4) Three
102. How many meiosis and mitosis required for formation of *polygonum* type of embryo sac?
(1) 1 and 1 (2) 1 and 2
(3) 1 and 3 (4) 1 and 4
103. Main mineral required for growth of pollen tube is :-
(1) B (2) Ca (3) Mn (4) Cu
104. In a typical angiospermic ovule :-
(1) Body is curved like horse shoe
(2) The body rotates by 180°
(3) Both body and embryo sac are curved
(4) Funicle is long and coiled around body

105. Coconut milk is :-
 (1) Cellular endosperm
 (2) Cellulosic endosperm
 (3) Free nuclear endosperm
 (4) One and three both
106. (i) Hydrophily - 300 genera
 (ii) Protandry - *Salvia*
 (iii) Lotus - Anemophily
 (iv) Pollen pistil interaction - Dynamic process
 How many above matches are incorrect?
 (1) Four (2) Three
 (3) Two (4) One
107. Ploidy of endosperm will be _____ if the male and female parents are hexaploid and tetraploid respectively :-
 (1) 8x (2) 7x (3) 16x (4) 10x
108. Select the odd one about pollination.
 (1) Elephant (2) Ant
 (3) Lemur (4) Bat
109. How many nuclei participate in double fertilisation?
 (1) Five (2) Four
 (3) Three (4) Ten
110. Pollen grains of *Vallisneria* are protected from wetting by :-
 (1) Covering of cellulose
 (2) Covering of sporopollenin
 (3) Covering of pecto cellulose
 (4) Covering of mucilage
111. Pick the correct one about hydrophily :-
 (1) Water lily (2) Water hyacinth
 (3) Lotus (4) *Hydrilla*
112. Which one of the following is not an out breeding device :-
 (1) Herkogamy (2) Dichogamy
 (3) Dioecy (4) Monocliny
113. Caruncle is found in :-
 (1) *Calotropis* (2) *Litchi*
 (3) *Ficus* (4) Castor
114. Mango, Peepal, *Commelina*, *Oxalis*, *Viola*
 How many plants from above list show assured seed set even in absence of pollinators :-
 (1) Four (2) Three
 (3) Two (4) One
115. Consider the following :-
 (i) Submerged
 (ii) Root less
 (iii) Monoecious
 (iv) Epihydrophily
 How many features are present in *vallisneria*?
 (1) Four (2) Three
 (3) Two (4) One
116. Find the odd one about pollination :-
 (1) Coconut (2) Mulberry
 (3) Date palm (4) Cotton
117. Ploidy of endosperm in mango is :-
 (1) n (2) 2n (3) 3n (4) 4n
118. Winged pollen grains are found in :-
 (1) *Pinus* (2) *Cycas*
 (3) Mango (4) *Calotropis*
119. When growth of the receptacle takes place inside the ovary in some flowering plants then it is called :-
 (1) Anthophore (2) Gynophore
 (3) Carpophore (4) Androphore
120. A such plant, which is perennial, but known as monocarpic :-
 (1) *Capsella* (2) *Mangifera*
 (3) *Agave americana* (4) *Citrus*
121. When embryo is directly formed other than female gametophyte, without meiosis and fertilization then is called :-
 (1) Apospory
 (2) Diplospory
 (3) Adventive embryony
 (4) None of the above

- | | |
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| <p>122. When development of pollen grain takes place on stigma then it is known as :-
 (1) Ex-situ
 (2) In-situ
 (3) In-vivo
 (4) Both (1) and (3)</p> <p>123. In some families of angiosperm, pollen grains of pollen sacs attached together to form a bag like structure is called :-
 (1) Massulae
 (2) Compound pollens
 (3) Pollinium
 (4) Polyspory</p> <p>124. 100 microspores in <i>Capsella</i> are formed by how many microspore mother cell?
 (1) 400 (2) 100
 (3) 50 (4) 25</p> <p>125. Germ pore in pollen grains of angiosperm are formed in :-
 (1) Intine
 (2) Exine
 (3) Both
 (4) None of the above</p> <p>126. In anther of angiosperm, cells of tapetum are :-
 (1) Haploid (2) Diploid
 (3) Polyploid (4) All of the above</p> <p>127. During meiotic division in microspore mother cell, division of cytoplasm is called :-
 (1) Cytomixis
 (2) Apomixis
 (3) Amphimixis
 (4) Cytokinesis</p> <p>128. The process of formation of pollen grains in anther is called :-
 (1) Microgametogenesis
 (2) Microsporogenesis
 (3) Endoscopic
 (4) Exosporic</p> | <p>129. If an organism is formed from a gamete without fertilization then it is called :-
 (1) Apogamy
 (2) Apospory
 (3) Parthenocarpy
 (4) Parthenogenesis</p> <p>130. Which of the following is vegetatively propagated through the root?
 (1) Ginger
 (2) <i>Chrysanthemum</i>
 (3) Sweet potato
 (4) Potato</p> <p>131. Slip is a special type of bud, which is present on which part of the plant?
 (1) Stem (2) Leaves
 (3) Tubers (4) Roots</p> <p>132. Clone is group of individuals of same type, which is obtained through :-
 (1) Self pollination
 (2) Cross pollination
 (3) Vegetative propagation
 (4) Hybridization</p> <p>133. Scion is a scientific term, which is used in relation to :-
 (1) Embryology
 (2) Agamospermy
 (3) Layering
 (4) Grafting</p> <p>134. In angiosperms, the process of pre-pollination development occurs in :-
 (1) Megasporangia
 (2) Sporophyll
 (3) Microsporangia
 (4) None of the above</p> <p>135. In angiosperm, dehiscence of anther takes place with the help of :-
 (1) Stomium
 (2) Tapetum
 (3) Middle layer
 (4) Endothecium</p> |
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136. Consider about the following plants ancestors :

- (a) Tracheophyte ancestors
- (b) Psilophyton
- (c) Chlorophyte ancestors
- (d) Rhynia type plants

Which of the following is correct sequence of the evolution of these plants ancestors?

- (1) c, d, a & b (2) c, a, b & d
- (3) c, a, d & b (4) a, c, d & b

137. Which of these is correct chronological sequence of plants in increasing evolutionary trend?

- (1) Seed ferns → Progymnosperms → Cycads
- (2) Progymnosperms → Seed ferns → Dicotyledons
- (3) Seed ferns → Cycads → Progymnosperms
- (4) Progymnosperms → Cycads → Seed ferns

138. Which of the following is not evolved from the sauropsid thecodont ancestor?

- (a) Tuataras (b) Birds
- (c) Mammals (d) Crocodiles
- (e) Dinosaurs
- (1) a and c (2) b and d
- (3) d and e (4) b, d and e

139. Select the correct evolutionary sequence of mammals :

- (1) Sauropsids → Pelycosaurus → Therapsids
- (2) Synapsids → Therapsids → Mammals
- (3) Pelycosaurus → Sauropsids → Mammals
- (4) Pelycosaurus → Synapsids → Mammals

140. Which of the following is mismatched pair of animals and its related information :

- (1) *Latimeria* (Coelacanth) - A living fossil
- (2) *Archaeopteryx* - A missing link
- (3) *Tyrannosaurus rex* - A large herbivorous dinosaur
- (4) *Pteranodon* - A flying dinosaur

141. Match the human ancestors of column A and their brain capacities of column-B and select the correct answer :

Column-A	Column-B
(A) <i>Homo erectus</i>	(i) 1650 cc
(B) <i>H. habilis</i>	(ii) 900 cc
(C) <i>H. sapiens neanderthalensis</i>	(iii) 1400 cc
(D) <i>H. sapiens fossilis</i>	(iv) 650-800 cc

Codes:

- (1) A-ii, B-iv, C-iii, D-i
- (2) A-ii, B-iv, C-i, D-iii
- (3) A-iv, B-ii, C-iii, D-i
- (4) A-iv, B-ii, C-i, D-iii

142. Which of the following ancestor is known to made prehistoric cave art?

- (1) *H. sapiens fossilis*
- (2) *H. sapiens neanderthalensis*
- (3) *H. erectus*
- (4) *H. habilis*

143. Which of following is not related with the *Australopithecus*?

- (1) Taung baby
- (2) Handyman
- (3) Lucy
- (4) They lived in east African grasslands about 2 mya

144. Which of the following arose about 75,000-10,000 year back (i.e. during the ice age)?

- (1) Neanderthal man
- (2) Modern man
- (3) *Homo habilis*
- (4) Java ape man

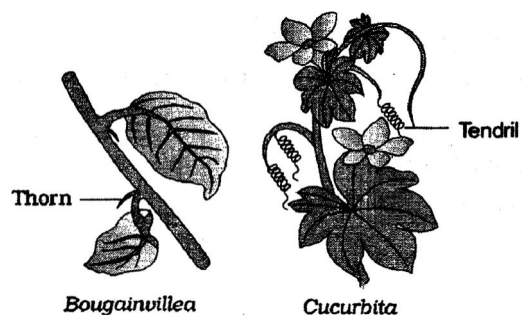
145. Which of the following is a mismatched pair of human ancestor and their characteristics?

- (1) *Homo habilis* - first tool maker
- (2) *Homo erectus* - first fire user
- (3) Neanderthal man - Herbivore ancestor
- (4) Cromagnon man - Closest ancestor of modern day man

146. Select a placental mammal which shows the parallelism with Tasmanian wolf?
- (1) Bobcat (2) Wolf
(3) Lemur (4) Flying squirrel
147. Which of the following is represented between Mole and Marsupial mole ?
- (1) Parallel evolution
(2) Divergent evolution
(3) Adaptive radiation
(4) Natural selection
148. Which of the following evidence is provided by Darwin's finches of Galapagos island :
- (1) Biogeographical evidence
(2) Evidences from branching descent
(3) Evidences from divergent evolution or adaptive radiations
(4) All of these
149. Select a correct statements :
- (1) Neanderthal man lived in south Africa
(2) Agriculture came around 18000 year back
(3) Skull of modern man resembles more closely with baby chimpanzee
(4) Cannibalism was common among *H. sapiens fossilis*
150. Which of the following extinct ancestor essentially ate fruits and hunted with stone weapons:
- (1) *Homo habilis*
(2) *Australopithecus*
(3) *Ramapithecus*
(4) *Dryopithecus*
151. Who used the term survival of fittest first time?
- (1) Herbert Spencer
(2) Charles Darwin
(3) J.B.de Lamarck
(4) Hugo de Vries
152. Which of the following scientists is known to believed that first form of life come from the nonliving molecules?
- (1) Louis Pasteur and Miller
(2) Operin and Haldane
(3) Reddi and Spallanzani
(4) Darwin and Lamarck
153. Which of following is incorrect about the *H. erectus*?
- (1) They probably did not ate the meat
(2) They has large brain capacity around the 900 cc
(3) They were first fire user
(4) They evolved from the *H. Habilis*
154. Which of following is incorrect about Miller's experiment:
- (1) He circulated H_2 , NH_3 , CH_4 and water vapour
(2) He maintained $800^\circ C$
(3) He obtained glutamic acid
(4) He used high voltage electrodes
155. The sequence of origin of life could be :
- (1) Organic materials $\rightarrow$ inorganic materials $\rightarrow$ colloidal aggregate $\rightarrow$ eobiont $\rightarrow$ cell
(2) Inorganic materials $\rightarrow$ organic materials $\rightarrow$ colloidal aggregates $\rightarrow$ eobiont $\rightarrow$ cell
(3) Inorganic materials $\rightarrow$ organic materials $\rightarrow$ eobiont $\rightarrow$ cell $\rightarrow$ colloidal aggregate
(4) Organic materials $\rightarrow$ inorganic materials $\rightarrow$ eobiont $\rightarrow$ cell $\rightarrow$ colloided aggregate
156. Information molecule to get evolved first on the primitive earth was :
- (1) Protein (2) DNA
(3) RNA (4) All of these
157. Fitness according to Darwin refers to :
- (1) Reproductive fitness
(2) Physiological fitness
(3) Spiritual fitness
(4) None of the above
158. Diversity of living organisms is due to :
- (1) Instant changes
(2) Polyploidy
(3) Long term evolutionary changes
(4) Short term evolutionary changes

159. Separate the following into homologous and analogous organs.
- Sweet potato & Potato
 - Eyes of Octopus & birds
 - Filippers of penguins and dolphins
 - Hearts of different vertebrate.
 - Forelimbs of Whales, Bat and Cheetah. The correct option is :
- Homologous organs ; Analogous organs
- I, II, III ; IV, V
 - IV, V ; I, II, III
 - I, II ; III, IV, V
 - I, II, V ; IV, III
160. In England in 1850s, i.e., before industrialisation set in, it was observed that there were more white-winged moths on trees than dark-winged or melanised moths. However, in the collection carried out from the same area, but after industrialisation, i.e., in 1920, there were more dark-winged moths in the same area, i.e., the proportion was reversed. Predict the possible reason for such change.
- Natural selection
 - Artificial selection
 - Conditional selection
 - Divergent selection
161. Choose the correct statements.
- Law of embryonic development was given by Von Baer.
 - Recapitulation theory was proposed by Spemann.
 - Haeckel theory states that 'Ontogeny repeats phylogeny'.
 - Haeckel theory and biogenetic law were proposed by the same person.
- The correct combination is :
- I and II
 - II and III
 - I, III and IV
 - I, II, III and IV
162. Evolution is not continuous. It is a Jerky and a discontinuous process. This is the punch line of
- Natural selection theory of evolution
 - Theory of acquired character
 - Mutational theory of evolution
 - Synthetic theory of evolution
163. There is a population in which more individuals are acquiring both peripheral character values other than the mean character value. This will be example of:
- Stabilising selection
 - Disruptive selection
 - Directional selection
 - All of these
164. 'Population tends to increase geometrically, while food supply increases arithmetically'. This concept was put forward by:
- T.R. Malthus
 - Stuart Mill
 - Charles Darwin
 - Adam Smith
165. Which of the following features are related with the modern theory of evolution ?
- Genetic and chromosomal mutation.
 - Genetic recombination and natural selection.
 - Reproductive isolation
- The correct combination is :
- I and II
 - II and III
 - I and III
 - I, II and III
166. In human beings, vestigial organs are :
- Wisdom teeth, coccyx, vermiform appendix, nail, eyelid
 - Wisdom teeth, coccyx, vermiform appendix, pancreas, elbow joint
 - Wisdom teeth, coccyx, vermiform appendix, nictitating membrane, auricular muscles
 - Coccyx, wisdom teeth, nail, auricular muscles
167. Bird with average sized wing survived in the severe storm but the short winged and long winged birds died. It shows:
- Stabilising selection
 - Gene flow
 - Diversifying selection
 - Founder effect

168. Identify the phenomenon in which the members of a species do not interbreed with the members of other species or same species :
- Natural selection
 - Geographical isolation
 - Mutation
 - Reproductive isolation
169. Forelimbs of cat, lizard used in walking ; forelimbs of whale used in swimming and forelimbs of bats used in flying are an example of
- Analogous organs
 - Adaptive radiation
 - Homologous organs
 - Convergent evolution
170. The wings of a bird and the wings of an insect are :
- Phylogenetic structures and represent divergent evolution
 - Homologous structures and represent convergent evolution
 - Homologous structure and represent divergent evolution
 - Analogous structures and represent convergent evolution
171. Industrial melanism is an example of:
- Mutation
 - Neo Lamarckism
 - Neo Darwinism
 - Natural selection
172. During 1952, J. Lederberg and E. Lederberg made a replica plating experiment by using an antibiotic on bacterial colonies. What was the finding of this experiment :
- Mutations are directed in nature.
 - Mutations are random and preadaptive.
 - Mutations are preadaptive but not random.
 - Mutations are neither random nor preadaptive.
173. Which of the following is correct about evolution A - Universe originated about 20 million years ago. B - Universe is formed by a galaxy, known as "milky way". C - Considering size of universe, earth is indeed a speck. D - Big-bang theory attempts to explain us the origin of solar system.
- Only A
 - Only B
 - Only A, B and C
 - Only C
174. Which of following is incorrect about the Louis pasteur
- He demonstrated life comes only from pre existing life forms.
 - He found that new living organisms could arose from killed yeasts in some flasks, open to air.
 - He supported the theory of abiogenesis.
 - He used swan necked flask in his experiment.
175. Select the incorrect statement :
- Miller obtained glutamic acid like amino acids in his experiment.
 - In other simulation experiments by other worker they observed formation of sugars, fats, pigments etc.
 - Miller circulated the hydrogen, ammonia and methane with a ratio of 2 : 1 : 2.
 - He maintained a higher temperature of 800°C.
176. The given figure shows an example of



- homologous organs
- convergent evolution
- divergent evolution
- both (1) and (3)

177. Select the statement which best explains the structures which develop as a result of divergent evolution:

- (1) Organs having no function but had important function in ancestors.
- (2) Organs appeared only in embryo.
- (3) Organs with functional similarity but anatomical dissimilarity.
- (4) Organs with anatomical similarity but functional difference.

178. Select an odd set of organs :

- (1) Thorns of *Bougainvillea* and tendrils of *Cucurbita*.
- (2) Flippers of Penguins and Dolphins.
- (3) Heart of frogs and human heart
- (4) Brain of different vertebrates.

179. Which of the following is not an example of anthropogenic evolution ?

- (1) Evolution of DDT resistant varieties of insects
- (2) Evolution of antibody resistant varieties of pathogens
- (3) Evolution of herbicide resistant varieties of weeds against crop plants
- (4) Evolution of long necked Giraffe in Africa.

180. Select the process by which man has obtained many breeds of plants and animals for agriculture, horticulture, food and security

- (1) Natural selection
- (2) Random breeding
- (3) Artificial breeding
- (4) Both (1) and (2)

